Nursing Practice Guideline

Regional Analgesia (Acute Pain)

Reference #: NPG14

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Regional Analgesia (Acute Pain). It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

For the purposes of this guideline, regional analgesia refers to the injection and infusion of local anaesthetic into a targeted space in the peripheral or central nervous system to provide pain management.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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1.0 Infection Prevention Principles

Observe the five moments for hand hygiene. Follow aseptic technique when managing regional analgesia lines and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.

2.0 General Management

Relevant to all regional therapies whether intraTHECAL, epidural, plexus or specific nerves.

2.1 Medical Requirements

All regional analgesia orders are written by the anaesthetist and must be documented on the appropriate chart. An anaesthetist orders the starting rate.

2.2 Safety Information

Pump: A specific rate-limited infusion pump, dedicated to regional analgesia must be used to regulate the infusion.(1)

Lines and Catheters:
Connection of local anaesthetic infusions to the regional catheter is only to be performed by:

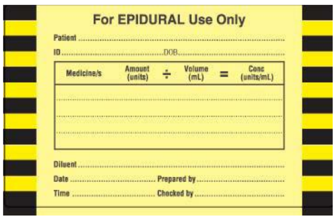
- An anaesthetist / anaesthesia registrar, or
- The medical practitioner who inserted the catheter / including specialty trainee, or
- Acute Pain Service nursing staff.
- Recovery staff in PACU

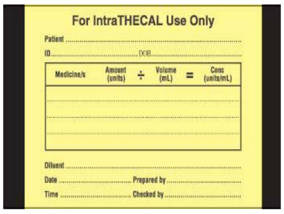
Neurological complications can occur following injection / infusion of neurotoxic agents such as alcohol(1) chlorhexidine and the preservatives used in many drugs prepared for intravenous use. To decrease the risk of this occurring:

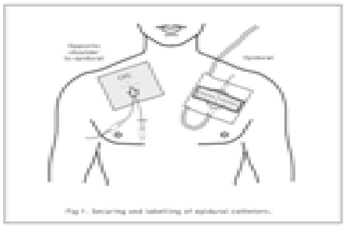
- All catheters/ lines must be labelled to identify the route of administration,(2)
- Labels are available for Epidural infusions, IntraTHECAL infusions, and regional infusions,
- All catheters must be labelled at filter site and on the infusion bag/ cassette with a yellow 'route of administration' label,(2)
- A dedicated yellow regional infusion line without any injection ports must be used for all regional infusions,(1)
- The rubber seal on infusion bags must NOT be swabbed with solutions containing alcohol or chlorhexidine(1) when changing bags. Ropivacaine 0.2% 200ml bags and Ropivacaine 0.2%/ Fentanyl 4mcg/ml 200ml bags are presented in sterile packaging with a sterile cap. The port requires no swabbing; use a non-touch technique to remove the cap and spike a new bag.
- Epidural, IntraTHECAL and Regional infusion **lines are NOT** to be changed by nursing staff.
- The set-up is left intact and not disconnected unless at the specific orders of the Acute Pain Service (APS).

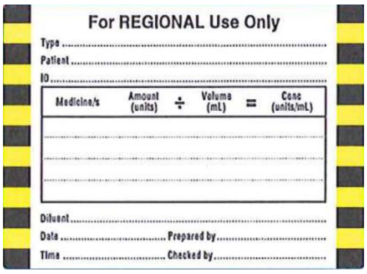
Disconnection: If a regional catheter becomes unexpectedly disconnected, wrap and secure the ends in sterile gauze using non-touch technique and contact the Acute Pain Team, or after hours, the on-call Anaesthetist.
Regional catheters should be secured to the opposite shoulder from a Central Venous Catheter.

Date and time of catheter commenced must be documented on the route of administration label.
Fluid bags/bottles for infusion labelled by hospital pharmacy or manufacturer DO NOT require additional labelling (i.e., additive label).









Metaraminol Bitartrate (Aramine™) and Naloxone (Narcan) must be available on the ward to treat hypotension and / or respiratory depression.

- Intravenous access must be present during the regional infusion and for six (6) hours post catheter removal.

Removal of Regional Catheters:

- Regional catheter removal is at the discretion of the Acute Pain Service,
- The timing of removal of all regional catheters should be as per Acute Pain Service Anticoagulation protocol [(1)],
- The specified time for **ceasing** (turning off) regional infusion and giving oral analgesia is indicated on the front of the Regional Analgesia Chart.

Leaving the Ward:
A patient with a regional infusion insitu must be accompanied by a nurse escort when away from ward. If the patient is non-compliant, the infusion is to be discontinued and:

- OPH: prescribing Anaesthetist contacted
- SCGH: Acute Pain Service or after-hours anaesthetist contacted.

2.3 Nursing Management

Urinary Catheter: An indwelling catheter is necessary for **all epidural and intraTHECAL infusions** and those regional infusions likely to affect bladder function (e.g. lower lumbar sacral area) and must be inserted immediately pre / post operatively.

- The urinary catheter should remain insitu until sensory and motor functions have returned completely.
- Monitor **urine output** on Fluid Balance Chart.

Infusion fluids:

- Available through Pharmacy.
- Fluid bags/bottles for infusion labelled by hospital pharmacy or manufacturer DO NOT require additional labelling (i.e. additive label).

Check at the beginning of each shift (8 hourly):

- Infusate; correct drug, correct concentration, correct route, as per the infusion order
- Pump program as per the prescription.
- Document ‘checked and correct’ on the inside of the Regional Analgesia Chart (MR810).
- Catheter insertion site and document:
    - Swelling
    - Bleeding
    - Signs of infection
    - Tenderness
    - Dressing intact
    - Excess fluid leakage (slight ooze or old blood is normal),
    - Catheter displacement

Observations:

Due to the potential for local anaesthetic to produce **motor block** (in addition to sensory block), assess **4 hourly** and document on the Regional, Epidural or IntraTHECAL Analgesia Chart. For the duration of an epidural or intrathecal infusion:

- Assess: sensory block level using the dermatome chart and motor block using the Bromage scoring scale.
- In the presence of motor block, perform pressure area care as appropriate.
- Do not mobilise patient until lower limb Bromage score = 0 and muscle strength has returned
- Report any abnormal signs and increasing motor block: OPH: Anaesthetist/ on-call Anaesthetist.

SCGH: Acute Pain Service or Anaesthetic Registrar on-call

- **The cumulative volume of the infusion must be recorded 4 hourly on the Regional, Epidural or IntraTHECAL Analgesia Chart**

Summary of observations:

Observations Required	**Documentation**	**Frequency**
-Sedation score -Nausea score -Functional activity score (FAS) - Motor block (Bromage) - Sensory block level -Volume infused (mls)	**Regional, Epidural or IntraTHECAL Analgesia Chart**	**4 hourly** for the duration of the infusion and as condition warrants. **After clinician bolus, assess sensory block and motor block and record at 15 mins and 30 minutes**
Other observations: -Temperature, BP, pulse, respiratory rate, level of consciousness, oxygen saturation and pain score	**Observation and Response (ORC) Chart** **ICU Flow Chart**	**4 hourly** for the duration of the infusion and as condition warrants. **Post clinician bolus assess and record BP at 15 mins and 30 minutes**
Insertion site assessment	**Regional, Epidural or IntraTHECL Analgesia Chart**	8 hourly for the duration of the infusion (and as condition warrants)

<u>Cease infusion; perform sensory assessment and contact Acute Pain Service immediately if patient experiences:</u>

- Tingling, numbness in arms, tongue or mouth,

- Increasing sedation, seizures
- Any pain at epidural / intrathecal catheter site,
- Profound or bilateral motor block or paraesthesia,

Epidural haematoma or abscess may result in permanent neurological damage if not treated immediately.

3.0 Regional Infusions

Includes Paravertebral, Erector Spinae, Lumbar Plexus, Brachial Plexus, Interscalene, Supraclavicular, Femoral Nerve, Adductor canal and Sciatic Nerve continuous infusions.

3.1 Medical Requirements

Regional analgesia orders are written by the anaesthetist and must be documented on the Regional Analgesia Chart.

3.2 Safety Information

The **route of administration** must be identified and labelled on all catheters / lines.

Nursing staff DO NOT administer PRN bolus doses to patients with plexus / nerve catheters.
An anaesthetist may prescribe either:
A continuous infusion and / or

- A patient-initiated bolus, OR
- A pump-initiated bolus

Where a Patient Controlled Regional Analgesia (PCRA) infusion is ordered a bolus dose may ***only*** be administered by the patient using the patient-controlled handset.

Pump Initiated Boluses
A ***'Pump Initiated Bolus'*** (labelled **'PIEB'**) together with a continuous infusion may be prescribed by the anaesthetist for non-neuraxial regional catheters. Regional techniques which may be used with pump-initiated boluses include (but are not limited to) rectus sheath catheters, TAP catheters, thoracic paravertebral catheters, lumbar plexus blocks and wound infusion catheters.
OPH: ambit® Infusion Pump
SCGH: The Sapphire regional infusion pump will be programmed by PACU staff. Boluses will be delivered automatically by the infusion pump at intervals prescribed by the anaesthetist.
The anaesthetist may request a delay in commencement of therapy. In this instance, the pump display will read 'KVO' until the first bolus is administered.
The rate and boluses are fixed in this therapy. No program changes are permitted by ward nursing staff.
***NOTE: For the duration of bolus delivery the pump will state a rate of 125ml/hr. Once the bolus has been delivered the pump will revert to the background infusion rate.

<u>Patients with regional blocks affecting their lower limbs must have motor function assessed prior to ambulation. Full motor function must be present prior to standing/ ambulation.</u>

- Patients should be observed for signs of local anaesthetic systemic toxicity (LAST). If any signs of LAST are observed numbness and tingling around the mouth
- irritability / restlessness / yawning
- altered conscious state

Immediately contact the Acute Pain Service or after-hours anaesthetist and initiate the actions outlined by the Observation and Response Chart (ORC) if required.

Patients who have received a regional block or infusion should have attained full motor function prior to discharge. If there is ongoing motor block, please contact the:

- OPH: prescribing Anaesthetist/ on call anaesthetist
- SCGH: Acute Pain Service or after-hours anaesthetic registrar.

3.3 Nursing Management

Care of Patients in Post Anaesthetic Care Unit (PACU)

- All patients transferred from general ward areas to Post Anaesthetic Care Unit (PACU) for insertion of plexus / nerve infusions:
  - Must be escorted by a nurse and have clinical handover completed.
  - Core physiological observations at commencement of infusion and at 15 minutes should be documented on the Observation and Response Chart
  - Must remain in PACU for a minimum of 15 minutes following insertion of the catheter and commencement of the infusion.
- All patients transferred from within the peri-operative suite post insertion of plexus / nerve infusions must remain in the PACU for a minimum of 30 minutes following administration of the block to allow assessment of the patient sedation, haemodynamic and respiratory status.
- Patients must have a set of core physiological observations completed 30 minutes post insertion (this may be done in either PACU or the ward area).
- Assess blood pressure, pulse, temperature, and pain score a minimum of 4 hourly (4/24) and document using the Observation and Response Chart
- Assess functional activity (FAS) scores 4 hourly (4/24) for duration of infusion, document using the Regional Analgesia Chart.
- Where a patient is administering their own bolus dose i.e., Patient Controlled Regional Analgesia (PCRA), extra observations are not required. Patient controlled bolus doses are small and therefore unlikely to cause complications.

4.0 Epidural - Local Anaesthetic +/- Opioid (e.g. Fentanyl) Infusions

4.1 Medical Requirements

Epidural infusions must be prescribed on the Epidural Analgesia Chart MR810.2 by an anaesthetist.

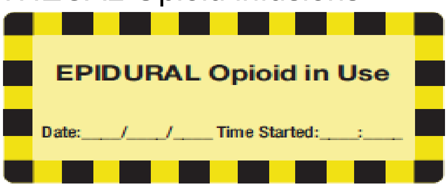
Ropivacaine dose **should not exceed** 400mcg/kg/hour. Lower doses should be prescribed in the context of renal disease, poor nutritional status and lowered seizure threshold.

4.2 Safety Information

For epidural infusions containing an opioid

No additional opioids should be given by any other route unless ordered by an anaesthetist.

An 'EPIDURAL Opioid in Use' sticker must be affixed to each of the patient's medication charts The Acute Pain Service does not use infusions of plain Intrathecal or Epidural opioids. Please refer to SCGH NPG Intrathecal Drug Delivery for guidelines and management of IntraTHECAL Opioid infusions.



Following commencement of an epidural infusion in Post Anaesthetic Care Unit (PACU), the patient is to remain in the PACU for a minimum of 30 minutes for observation.

Rates and Boluses:

- Rate may be changed by nursing staff within the specified rate range as per order.
- An anaesthetist may prescribe either
    - A staff-initiated bolus of epidural infusate, OR
    - A patient-initiated bolus of epidural infusate.
- Staff OR patient-initiated bolus may be given if ordered on the Regional Analgesia Chart.
- Pump initiated boluses should not be prescribed for epidural or intraTHECAL infusions.

Bolus of local anaesthetic in Post Anaesthetic Care Unit (PACU):

- Record blood pressure and heart rate every 5 minutes for 15 minutes
- Patients are to remain in PACU for a minimum of 30 minutes following an epidural bolus of local anaesthetic.

Observations post bolus of epidural solution: Local anaesthetics block transmission to the sympathetic nervous system fibres responsible for constrictor tone in arteries and veins [3]. This may result in vasodilation and hypotension. The patient must therefore be monitored following a bolus:

- o At 15 minutes: check blood pressure,
- o At 30 minutes: check blood pressure, pain (rest and movement), sensory block, sedation, conscious level and functional Activity Score (FAS).

Intravenous Cannula must remain insitu for the duration of the epidural infusion and for a minimum of 6 hours post cessation of infusion.

<u>Block height:</u>

If block height by dermatome chart is above **T4** (nipple level) with any **one** of the following compromising clinical features including:

- Systolic blood pressure < 90 mmHg
- Heart rate < 50 bpm
- Respiratory rate < 8 breaths/min
- Sedation score > 3
- Change in sensation on the inner aspect of the arm and/or unable to curl and extend fingers to grasp both of your hands = block height higher than **T3**.

ACTION: STOP epidural infusion. Contact the Acute Pain Service or the Anaesthetist on call (p4823).

If block height by dermatome chart is above **T4** (nipple level) with **no** compromising clinical features:

ACTION: Continue epidural at the SAME rate.

- NO boluses and NO rate increases.

Hourly observations for 4 hours, to ensure the block height remain stable.

Managing problems:

Escalation responses as per the Observation and Response Chart apply for all observations during an epidural infusion. If an observation falls within a shaded area, the actions specified for that colour must be initiated.

The following observations require medical review:

- Respiratory rate < 9 breaths per minute,
- Heart rate less than 50 beats per minute,

- Systolic blood pressure less than 90 mmHg or as per specific order as documented on front of Epidural Analgesia Chart (See 'HYPOTENSION').

Cease infusion; perform sensory assessment and contact Acute Pain Service immediately if patient experiences:

- Tingling, numbness in arms, tongue or mouth,
- Increasing sedation, seizures
- Any pain at epidural catheter site,
- Profound or bilateral motor block or paraesthesia.

Epidural haematoma or abscess may result in permanent neurological damage if not treated immediately.

4.3 Nursing Management

Oxygen: Deliver by Hudson mask (6L/min), Airvo or nasal prongs (2L/min) as ordered.

Dermatome level:

- Determine the extent of sensory blockade (dermatome distribution) using ice 4 hourly (4/24) for the first 24 hours following insertion of epidural catheter.
- After 24 hours, check block level 4 hourly and as condition warrants. Assess dermatome level in response to patient complaints of pain, changes in other observations, or if any other concerns re: epidural.
- Check dermatome levels prior to, and 30 mins after a bolus.

Motor Block: Record Bromage score 4 hourly and prior to standing or ambulation.

- **If Bromage score = 2 or 3, notify the Acute Pain Team or the on-call Anaesthetic Registrar.**
- Continue to monitor lower limb Bromage score until = 0 bilaterally.
- Do not ambulate patient unless lower limb Bromage score = 0 and muscle strength has returned

Assessment of core physiological observations:

- Assess temperature, blood pressure, pulse, respiratory rate, oxygen saturation, conscious state and pain score 4 hourly (4/24) for the duration of the infusion and as condition warrants. Observations are to be recorded on the Observation and Response Chart
- Epidural analgesia catheter insertion site must be observed and assessed 8-hourly for the duration of the infusion (document on the Epidural Analgesia Chart).

Nurse initiated bolus:

- A registered nurse / medication competent enrolled nurse may administer a bolus dose of epidural solution and alter the rate of the epidural infusion as prescribed on the Epidural Analgesia Chart.
- Two nurses (one of whom must be a RN) must remain at the bedside until the bolus dose has been delivered and the epidural infusion has been re-started at the desired rate
  - Prior to a bolus, check sensory blockade and blood pressure. If upper level of block is above T4, do not give bolus and contact Acute Pain Service.
  - Administer up to 4mls of the infusate as a bolus, as prescribed.
  - Following a bolus:
  - At 15 minutes check blood pressure
  - At 30 minutes check blood pressure, pain score and dermatome level.
  - If current epidural rate has been inadequate for analgesia, the infusion rate should be increased by 2 mls at the time of bolus.

  - If adequate analgesia is not attained after two (2) consecutive bolus doses, 30 minutes apart contact the Acute Pain Service or the Anaesthetist on-call.

Patient Controlled Epidural Analgesia (PCEA)

- Anaesthetists may prescribe bolus doses that are to be administered by the patient via a hand control. The hand control is for patient use only – Nursing staff DO NOT administer any bolus doses either by hand control or pump.
- The PCEA prescription/ order may include a continuous infusion and a limit to the amount that the patient can receive in one hour.

All observations and care is as above with the following exception:

- Observations by nursing staff are not required after a patient-initiated bolus dose.

Patient education should include use of the hand control (i.e., the hand control is to be used if pain present / should not be used if the patient is comfortable.)

5.0 IntraTHECAL Local Anaesthetic Infusion

5.1 Medical Requirements

IntraTHECAL infusions must be prescribed on the IntraTHECAL Analgesia Chart MR810.3 by an anaesthetist.
The solution must be Ropivacaine 0.1%.
Ropivacaine dose **should not exceed** 6 mg / hr
The Acute Pain Service does not use infusions of plain Intrathecal opioids. Please refer to SCGH NPG Intrathecal Drug Delivery for guidelines and management of IntraTHECAL Opioid infusions.

5.2 Safety Information

Any rate changes or boluses of IntraTHECAL infusate must be by an anaesthetist or anaesthetic registrar. The pump-initiated bolus mode is not to be used for IntraTHECAL infusions.
Following commencement of an IntraTHECAL infusion in the Post Anaesthetic Care Unit (PACU), the patient is to remain in the PACU for a minimum of 30 minutes for observation.

Bolus of local anaesthetic in Post Anaesthetic Care Unit (PACU):
Record blood pressure and heart rate every 5 minutes for 15 minutes
Patients are to remain in PACU for a minimum of 30 minutes following an IntraTHECAL bolus of local anaesthetic.

Intravenous Cannula must remain insitu for the duration of the IntraTHECAL infusion and for a minimum of 6 hours post cessation of infusion.

<u>Block height:</u>
If block height by dermatome chart is above **T4** (nipple level) with any **one** of the following compromising clinical features including:

- Systolic blood pressure < 90 mmHg
- Heart rate < 50 bpm
- Respiratory rate < 8 breaths/min
- Sedation score > 3
- Change in sensation on the inner aspect of the arm and/or unable to curl and extend fingers to grasp both of your hands = block height higher than **T3**.

ACTION: STOP IntraTHECAL infusion. Contact the Acute Pain Service or the Anaesthetist on call (p4823).

If block height by dermatome chart is above **T4** (nipple level) with **no** compromising clinical features:
ACTION: Continue IntraTHECAL infusion at the SAME rate.
- NO boluses and NO rate increases.

Hourly observations for 4 hours, to ensure the block height remain stable.

Managing problems:
Escalation responses as per the Observation and Response Chart apply for all observations during an epidural infusion. If an observation falls within a shaded area, the actions specified for that colour must be initiated.
The following observations require medical review:

- Respiratory rate < 9 breaths per minute,
- Heart rate less than 50 beats per minute,
- Systolic blood pressure less than 90 mmHg or as per specific order as documented on front of IntraTHECAL Analgesia Chart (See 'HYPOTENSION').

<u>Cease infusion; perform sensory assessment and contact Acute Pain Service immediately if patient experiences:</u>

- Tingling, numbness in arms, tongue or mouth,
- Increasing sedation, seizures
- Any pain at IntraTHECAL catheter site,
- Profound or bilateral motor block or paraesthesia.

Epidural haematoma or abscess may result in permanent neurological damage if not treated immediately.

5.2 Nursing Management

Oxygen: Deliver by Hudson mask (6L/min), Airvo or nasal prongs (2L/min) as ordered

Dermatome level:

- Determine the extent of sensory blockade (dermatome distribution) using ice 4 hourly (4/24) for the first 24 hours following insertion of IntraTHECAL catheter.
- After 24 hours, check block level 4 hourly and as condition warrants. Assess dermatome level in response to patient complaints of pain, changes in other observations, rapidly ascending block level or if any other concerns re: IntraTHECAL
- Check dermatome levels 30 mins after an anaesthetist administered bolus.

Motor Block: Record Bromage score 4 hourly and prior to standing or ambulation.

- **If Bromage score = 2 or 3, notify the Acute Pain Team or the on-call Anaesthetic Registrar.**
- Continue to monitor lower limb Bromage score until = 0 bilaterally.
- Do not ambulate patient unless lower limb Bromage score = 0 and muscle strength has returned

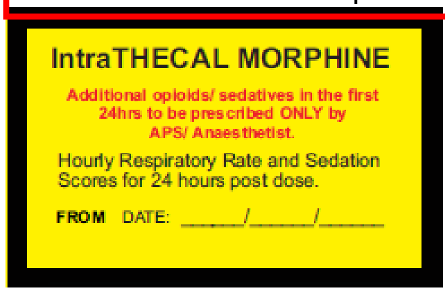
Assessment of core physiological observations:

- Assess temperature, blood pressure, pulse, respiratory rate, oxygen saturation, conscious state and pain score 4 hourly (4/24) for the duration of the infusion and as condition warrants. Observations are to be recorded on the Observation and Response Chart
- Assess blood pressure and pulse 15 minutes following an anaesthetist administered bolus
- IntraTHECAL analgesia catheter insertion site must be observed and assessed 8 hourly for the duration of the infusion (document on the Epidural Analgesia Chart).

6.0 Post-Operative Management of Intraoperative IntraTHECAL Morphine

6.1 Medical Requirements

The anaesthetist / surgeon will administer intraTHECAL morphine into the cerebrospinal fluid during the operation.
The dose and time of administration is recorded on the inside of the Anaesthesia Record. Medical staff should also complete and apply a yellow IntraTHECAL Morphine administration label to patient's Medication chart/s and Intravenous Opioid Analgesia



If the intrathecal dose of morphine is large (e.g. >200 micrograms) or patient has significant co-morbidity, the anaesthetist may request transfer to intensive care unit or high dependency unit for the first postoperative night.

6.2 Safety Information

Single dose intraTHECAL morphine (e.g. in doses of 100 to 300 micrograms) can provide effective postoperative analgesia for up to 24 hours.

Delayed respiratory depression is a potentially dangerous complication(3) as it may occur hours after i intraTHECAL injection of opioid when intraTHECAL morphine reaches respiratory centres in the medulla. Onset time in opioid naive patients may occur from 3.5 to 7.5 hours post dose. However this may occur later as the duration of intraTHECAL morphine is up to 24 hours. [(4)]

Intravenous access must remain in situ for 24 hours after injection.

intraTHECAL morphine will cause detrusor muscle relaxation, which may lead to urinary retention.(2)

6.3 Nursing Management

Patient may be nursed in a general ward area with the following care:

- Wake patient prior to assessment of respiration and sedation 1/24 for 24 hours after injection.
- Record hourly respiration rate and consciousness level on ORC chart

PLEASE NOTE: ***consciousness* level is being used to reflect patient's ***sedation*** levels following administration of intrathecal morphine. Please consider that you need to differentiate sedation due to intraTHECAL l morphine from change in consciousness level due to other causes.

- If respiratory rate or consciousness level falls within range for escalation, clinicians must respond to the abnormal physiological observations by following the ORC escalation and actions required.
- Record 4/24 blood pressure, pulse, temperature and pain score using the Observation and Response Chart
- Record functional activity scores (FAS) and nausea in the nursing notes.
- Administer oxygen at 2L/min via nasal prongs or 5L/min via facemask for 24 hours.
- If an indwelling catheter is not in-situ on return to ward, perform bladder scan.
- Bladder scans should then be performed as per SCGOPHCG Nursing Practice Guideline Bladder Management and Urinary Catheterisation for 24 hours.

7.0 Cessation of Regional Therapy

7.1 Medical Requirements

Cessation of regional therapy will be ordered by the anaesthetist / Acute Pain Service (APS) and documented on the front of the Regional Analgesia Chart

7.2 Safety Information

Haematoma is a rare but a potentially serious complication of regional catheter removal (1). Check if patient has received any anticoagulation therapy &/ or had coagulation profile within 24hrs before removing catheter.

Patients on Intravenous Infusion of heparin or other anticoagulation agents should be reviewed by Acute Pain Service prior to catheter removal.

Acute Pain Service anticoagulation guide		
Agent	**Waiting time for removal after dose**	**Next dose can be given**
SCI unfractionated heparin	6 hours	2 hours
Enoxaparin (clexane) low (prophylactic) dose (e.g. 0.5mg/kg)	12 hours	2 hours
(Enoxaparin (clexane) therapeutic dose (e.g. 1.5mg/kg)	22 hours	2 hours
Rivaroxaban (oral)(5)	18 hours	6 hours
Apixaban (oral)	12 hours	2 hours

7.3 Nursing Management

- Cease regional infusion at stated time and administer analgesia as prescribed. If analgesia adequate 3 hours later, check anticoagulation status as per Acute Pain Service policy then remove regional catheter.
- An indwelling urinary catheter must be inserted immediately pre/post operatively if the regional infusion is likely to affect bladder function (e.g. lower lumbar sacral area). The urinary catheter should remain insitu at least 2 hours following removal of regional catheter.

7.4 Procedural Information

Requirements

- Personal Protective Equipment
- IV pressure pad dressing
- Dressing pack
- Sterile saline solution
- 2% Chlorhexidine in Alcohol 70% swab stick
- Swab collection kit
- Sterile specimen container
- Sterile scissors

Procedure

1. Explain procedure to patient.
2. Perform hand hygiene. Position patient to expose catheter insertion site.
3. Removal of intraTHECAL/ epidural catheters can be optimised by the patient:
   a. Sitting forward with their neck flexed and body curved forward hugging a pillow, or
   b. Lying in the lateral position with knees drawn towards the chest and neck flexed.
4. Release the tape securing the catheter.
5. Perform hand hygiene and don gloves.
6. If redness, swelling or pus present at site, proceed as follows:
   a. Irrigate with sterile saline prior to swabbing to eliminate surface contaminants. Use Swab Collection Kit to obtain swab (lightly moisten swab with sterile saline if wound is dry) for Microscopy, Culture and Sensitivity (MC&S) of insertion site,
   b. Withdraw catheter and check that tip is intact,
   c. Clean around site with 2% Chlorhexidine in 70% Alcohol swab stick and allow to dry,
   d. Using sterile scissors cut the catheter 2 cm from tip,
   e. Clearly label and send both specimens to Microbiology for culture,
   f. Contact Acute Pain Service immediately to review patient.
7. If no signs of infection:

a. Gently withdraw catheter and check that tip of catheter is intact,
b. Clean area with 2% Chlorhexidine in 70% Alcohol swab stick and allow to air dry.

8. Place IV pressure pad dressing over site.

9. Discard gloves and perform hand hygiene.
10. Document date and time of removal of catheter, and whether tip intact on Regional Analgesia Chart.

Observe and document epidural, intraTHECAL or regional insertion site once daily until patient discharge. Notify Acute Pain Service if redness, swelling, exudate or pain develops at site.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010
- Work Health and Safety Act 2020

Policy:

- Health Mandatory Policy 0131/20 High Risk Medication Policy.
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Control manual -Aseptic Technique Policy.
- NMHS Patient Identification and Procedure Matching Policy
- SCGOPHCG Patient Identification
- NMHS Administration of S8 and S4R Medicines Policy and Procedure
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicine Policy
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicine Procedure
- SCGOPHCG Medicines Management Policy and Procedure

References

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Authorisation & Version Control

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<tr><td>Date First Issued:</td><td>August 2000</td><td>Last Reviewed:</td><td>December 2024</td><td>Review Date:</td><td>December 2027</td></tr>
<tr><td>Version No.</td><td colspan="5">11.0</td></tr>
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<tr><td>Standards Applicable:</td><td colspan="5">NSQHS Standards (Version 2):</td></tr>
</table>

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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